EE P 509 A Project Proposal

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Project Proposal Title:

Membership Inference Attacks on Knowledge Distillation: Empirical Evaluation and Mitigation on Healthcare Data

Problem Description

Component 1: Project Proposal

1. Problem Description

Knowledge distillation compresses a large "teacher" model into a smaller "student" by training it on soft output probabilities rather than hard labels. This project studies this setup using Texas-100X, a dataset of 925,128 patient records from 441 Texas hospitals, where the task is predicting surgical procedure codes from demographic and medical features. The sensitive asset is patient membership status in the training set. Knowing whether a specific person's record was used to train the model can reveal that they underwent a particular surgical procedure, which is protected health information carrying HIPAA liability and discrimination risk. Two attacker profiles are relevant: an external party with black-box API access to the deployed student, and a malicious insider with white-box access to the distillation pipeline. Both threats are enabled by a widely held assumption that distillation improves privacy because the student never sees raw training data. In reality, soft labels can carry confidence signals that reflect how thoroughly the teacher memorized individual records, and students trained on those signals can inherit or amplify that leakage. The teacher's accuracy gap on Texas-100X (61% train vs. 46% test) is the type of overfitting that membership inference attacks exploit. I was inspired by the 2026 Anthropic/OpenAI disclosure of industrial-scale distillation attacks to study this further.

2. Proposed Approach

This project tests whether knowledge distillation preserves, transfers, or amplifies membership inference vulnerability, and evaluates three targeted mitigations.

Methodology:

1. Train a teacher model on 50,000 Texas-100X records.

2. Distill a student model from the teacher's soft labels on the same training subset.
3. Apply three MIA attacks on both models: the shadow model attack (Shokri et al. 2017), the loss-based attack (Yeom et al. 2018), and LiRA (Carlini et al. 2022) as the current gold standard.
4. Compare attack AUC (area under curve) across teacher vs. student to determine whether distillation transfers privacy risk.
5. Apply and evaluate three mitigations from recent literature: (a) filtering out high-confidence training points before distillation, (b) adding a Bottleneck Projection layer to limit information transfer, and (c) applying NoNorm normalization.
6. Plot privacy-utility tradeoff curves: baseline teacher → student → mitigated student.

The main idea this project aims to deliver is a clear empirical answer to a question that surprisingly lacks one: does distilling a model actually reduce its privacy risk, or does it just move that risk somewhere less visible? Beyond the result itself, the evaluation pipeline will be structured to be reproducible so future work can build on it. One tradeoff is that the mitigations likely hurt accuracy to some degree, and how much will depend on how aggressively they're applied. One limitation is that the Texas-100X is a table-format healthcare dataset, and findings here will not automatically transfer to image classifiers or language models (that boundary will be addressed in the report). On an ethical note, the dataset is used strictly for research and no attempt will be made to re-identify any individual, and the hospital identifier (THCIC_ID) is excluded from all model inputs in line with the dataset's own usage guidelines.

Component 2: Project Plan

The final delivery will include a code repository with teacher/student training scripts, MIA attack implementations (LiRA, shadow model, loss-based), and mitigation code. It will also include figures showing privacy-utility tradeoff curves, MIA AUC comparisons across models, and ablation results for each mitigation.

Week	Goals
1	Submit proposal. Set up environment (PyTorch, ML-Privacy-Meter, SELENA). Download and verify Texas-100X.
2	Train teacher model. Confirm accuracy gap (~61% train / ~46% test). Run shadow model and loss-based MIAs on teacher.

3	Implement KD, train students, run the same MIAs. Compare teacher vs. student attack AUC.
4	Implement LiRA on both models. Begin mitigation 1 (filter high-confidence members before distillation).
5	Implement mitigations 2 and 3 (Bottleneck Projection, NoNorm). Re-run full MIA suite on mitigated students.
6	Run ablation study, generate tradeoff curves, begin writing report.
7	Finalize report, clean up code repository, write README, prepare for presentation.

References

1. Shokri, R., Stronati, M., Song, C., & Shmatikov, V. (2017). *Membership Inference Attacks Against Machine Learning Models.* IEEE S&P. arXiv | IEEE
2. Yeom, S., Giacomelli, I., Fredrikson, M., & Jha, S. (2018). *Privacy Risk in Machine Learning: Analyzing the Connection to Overfitting.* IEEE CSF. arXiv | PDF
3. Carlini, N., Chien, S., Nasr, M., Song, S., Terzis, A., & Tramer, F. (2022). *Membership Inference Attacks From First Principles.* IEEE S&P. arXiv | IEEE
4. Tang, X., Mahloujifar, S., Song, L., Shejwalkar, V., Nasr, M., Houmansadr, A., & Mittal, P. (2022). *Mitigating Membership Inference Attacks by Self-Distillation Through a Novel Ensemble Architecture (SELENA).* USENIX Security. Paper | Code
5. Cui, Z., Zhang, M., & Pei, J. (2025). *Membership Inference Amplification via Knowledge Distillation: Attacks and Mitigations.*arXiv:2505.11837
6. Jayaraman, B. et al. *Texas-100X Dataset.* GitHub

AI Note: I workshopped the title of the project a bit using Claude 4.6 Sonnet and also used AI to point me in the right direction of papers to read with relation to my topic of interest. The papers selected were from me based on the interest/relation to my project. All other writing is mine and mine only.